Sham Satish Thakare

Pre-Doctoral Researcher · Machine Learning, Distributed Systems & Computational Foundations

Capital One Systems Research · APCOER Pune (Savitribai Phule Pune University)

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RESEARCH INTERESTS

Foundations of Machine Learning & Reasoning: Reinforcement learning with verifiable rewards (RLVR), policy optimization (GRPO/PPO), uncertainty calibration, length confounding, and state-matched error recovery dynamics.
Fault-Tolerant Distributed Systems: State-machine replication (Raft, Paxos), dynamic quorum adaptation under network partitioning, linearizability verification, and tail latency minimization in modern C++20.
Mathematical & Quantum Foundations: Spectral graph theory, convex optimization, stochastic processes, quantum information foundations (density matrix formalism, unitary dynamics), and quantum circuit compilation complexity.

EDUCATION & HONORS

APCOER Pune (Savitribai Phule Pune University), Pune, India **01/2020 – 06/2024**

Bachelor of Technology (B.Tech.) in Artificial Intelligence and Data Science | **Cumulative GPA: 8.70 / 10.00** **Rank 1 (9.53 SGPA)**

- **Gold Medal in Mathematics (2024):** Awarded for highest academic rank across mathematical and quantitative curricula in the university cohort.
- **Academic Standing:** Final Year SGPA **9.53 / 10.00** (First Class with Distinction, Department Rank 1).
- **Mathematics & Theory Coursework:** Advanced Linear Algebra (Spectral Decomposition, SVD), Probability & Stochastic Processes, Optimization Theory, Discrete Mathematics & Graph Theory, Calculus & Differential Equations.
- **Computer Science Coursework:** Design & Analysis of Algorithms, Data Structures, Distributed Computing, Operating Systems, Machine Learning, Deep Learning, Computational Intelligence, Database Management Systems.

RESEARCH & PROFESSIONAL EXPERIENCE

Capital One Systems Research, Pune, India **06/2024 – Present**

Pre-Doctoral Researcher (Distributed Systems & Machine Learning) | Mentorship: **Yuri Shelke** (Distinguished Engineer)

- Formulated dynamic quorum adaptation algorithms for Raft consensus clusters (quorumshift), proving invariant quorum overlap ($Q_i \cap Q_j \neq \emptyset$) under dynamic replica re-weighting without violating state-machine linearizability.
- Developed an online latency-profiling and anomaly-filtering estimator eliminating p99 tail latency degradation under asymmetric network fault injection and non-uniform round-trip delays.
- Engineered an asynchronous, lock-free C++20 distributed benchmarking harness evaluating multi-node consensus throughput across heterogeneous network partitions and emulated cross-datacenter WAN topologies.

APCOER Pune (Savitribai Phule Pune University), Pune, India **06/2023 – 05/2024**

Undergraduate Researcher & Teaching Assistant | Advisors: **Prof. Sagar Gawande** & **Prof. Sunil Thakare**

- Investigated token predictive entropy behavior in autoregressive reasoning models, discovering strong completion-length confounding ($r = +0.486$) that undermines heuristic uncertainty weighting in reinforcement learning.
- Designed state-matched continuation evaluation protocols isolating nominal error self-correction from unconditional generative capacity under frozen structural covariates.

• Served as Teaching Assistant for undergraduate courses in *Data Structures, Algorithms, and Foundations of Machine Learning*, conducting tutorials and recitations for cohorts of 120+ engineering students.

PUBLICATIONS & MANUSCRIPTS UNDER REVIEW

• When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in Language Model Reinforcement Learning

Sham Satish Thakare. *Under Review at IEEE Transactions on Artificial Intelligence (IEEE TAI)*, 2024. [Code] [Research Note]
Proved empirical completion-length confounding (r = +0.486, p < 10^-6) in token predictive entropy, proving that confidence weighting systematically penalizes valid multi-step derivations in GRPO; conducted five-way matched ablation study on GSM8K/MATH demonstrating 0.00% Pass@1 gain over length-controlled baselines.

• AdaptiveReplica: Dynamic Quorum Adaptation and Failure-Aware Replica Selection in Distributed Consensus

Sham Satish Thakare, **Yuri Shelke** (Distinguished Engineer, Capital One). *Under Review at IEEE Transactions on Artificial Intelligence (IEEE TAI)*, 2024. Capital One Systems Research. [Code (C++20)]
Formulated online latency profiling algorithms in C++20 Raft consensus clusters that dynamically re-weight quorums, proving non-empty intersection invariants ($Q_i \cap Q_j \neq \emptyset$) under bounded clock drift Δ, eliminating p99 tail latency degradation under asymmetric network jitter and partitions.

• recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning

Sham Satish Thakare. *Under Review at 11th IEEE Special Session on Machine Learning on Big Data (IEEE BigData)*, 2024. [PDF Paper] [Code]
Formulated state-contrast continuation evaluation protocol ($D_{recovery} = E[\Delta_{recovery}] - E[\Delta_{control}]$) isolating true error recovery from baseline generative capacity under frozen structural covariates across reasoning model rollouts.

• MediRush-SafeAgent: Layered Runtime Enforcement for Healthcare-Commerce LLM Tool Execution

Sham Satish Thakare. *In Peer Review at Intelligence-Based Medicine (Elsevier) & SSRN Preprint (ID: 7446543)*, 2024. [SSRN] [Code]
Formulated multi-stage deterministic runtime policy interception and out-of-band schema validation barriers that decouple proposed calls from system mutations, guaranteeing 100% safe task completion and 0% policy violations.

PREPRINTS & WORKING PAPERS

• **StateShift: Tracking State-Dependent Reasoning Recovery Dynamics in Neural Sequence Models**: **Sham Satish Thakare**. *Technical Report & Preprint*, 2024. [PDF] [Code] — Evaluates continuation state matching under verifier perturbations and structural covariate controls.

• **Amortized Intervention Frontiers for Language-Model Reasoning: When Does Training Beat Search?**: **Sham Satish Thakare**. *Working Paper in Preparation for Transactions on Machine Learning Research (TMLR)*, 2024. [Code & Benchmarks] — Characterizes compute Pareto frontiers comparing guided test-time search versus post-training RL.

• **Predicting Reinforcement-Learning Plasticity of Intermediate Language-Model Checkpoints**: **Sham Satish Thakare**. *Technical Report & Diagnostic Probing Suite*, 2024. [Code] — Formulates gradient norm and feature drift diagnostic probes across model checkpoints, predicting reward plasticity prior to RL loops and reducing FLOPs by 18.7%.

THEORETICAL & METHODOLOGICAL CONTRIBUTIONS

• **Credit Assignment Geometry in Reasoning RLVR (Mathematical ML):** Derived analytical bounds for token predictive entropy $H(Y_t \mid Y_{<t}, X)$ in autoregressive reasoning sequences; proved empirical collinearity with sequence length ($r = +0.486$, $p < 10^{-6}$), demonstrating that heuristic uncertainty weighting acts as an implicit length penalty that degrades credit allocation on complex proofs; established 0.00% Pass@1 gain across multi-seed controls on GSM8K/MATH.

• **Quorum Overlap Invariants in Asymmetric State-Machines (Distributed Systems Theory):** Formulated dynamic quorum adaptation algorithms in C++20 Raft consensus clusters (quorumshift); proved that under bounded clock skew Δ and asynchronous network delivery, dynamically re-weighted voting quorums preserve non-empty majority intersections ($Q_i \cap Q_j \neq \emptyset$), strictly preserving linearizability while eliminating p99 tail latency degradation.

• **Quantum Information & Computational Foundations (Quantum Science & Computing):** Investigated multi-qubit density matrix formalism (ρ ■ 0, Tr(ρ)=1), von Neumann entropy S(ρ) = -Tr(ρ log ρ), and quantum state fidelity bounds; analyzed computational complexity separations (BPP ⊆ BQP ⊆ PSPACE) and query complexity of Grover search on structured Hamiltonians; studied stabilizer code projections and tensor network contraction methods for simulating distributed quantum state transitions.

• **Counterfactual State-Contrast Evaluation (Statistical Evaluation Methodology):** Formulated the state-contrast continuation evaluation metric (D_recovery = E[Δ_recovery] - E[Δ_control]) in recovery_eval, establishing rigorous counterfactual controls that isolate true reasoning self-correction from unconditional generative capacity under frozen structural covariates.

ACADEMIC LEADERSHIP & SERVICE

• **President, Social Volunteer Club**, APCOER Pune (2023 – 2024): Elected student head of university community outreach; directed educational mentorship in quantitative and digital skills for 200+ underprivileged high-school students, organized city-wide blood donation drives, and led campus sustainability campaigns.

• **Teaching Assistant**, APCOER Pune (2023 – 2024): Conducted weekly recitation sessions, algorithmic proof walkthroughs (graph traversal, dynamic programming, complexity classes), and PyTorch implementation labs for 120+ undergraduate engineering students in *Data Structures, Algorithms, and Foundations of Machine Learning*.

TECHNICAL SKILLS & SCIENTIFIC TOOLS

• **Pure & Applied Mathematics:** Linear Algebra (Spectral Decomposition, SVD, Rayleigh Quotients, Perturbation Bounds), Real Analysis (Metric Spaces, Lebesgue Measure), Abstract Algebra (Groups, Rings, Fields), Probability Theory, Stochastic Processes, Convex Optimization (KKT Conditions, Subgradient Methods), Discrete Mathematics, Graph Theory.

• **Quantum Computing & Information:** Density Matrix Formalism, Unitary Operators, Stabilizer Codes, Quantum Error Correction, Quantum Algorithms (Shor, Grover, Quantum Walks), Quantum Complexity (BQP, BPP, QMA), Qiskit, Cirq, PennyLane.

• **Machine Learning & AI:** Reinforcement Learning (PPO, GRPO, RLVR), Transformer Architectures, Statistical Learning Theory, PyTorch, JAX, Hugging Face Transformers, NumPy, SciPy.

• **Systems Programming & Infrastructure:** C++ (C++20 Multithreading, Lock-free Ring Buffers), Distributed Consensus (Raft), POSIX, Linux Kernel Internals, OpenTelemetry, Git, Docker, LaTeX.

ACADEMIC & RESEARCH REFERENCES

• **Yuri Shelke**, Distinguished Engineer, Capital One · *Systems Research Mentor & Co-author (Distributed Systems & Consensus)* · Contact available upon request

• **Prof. Sagar Gawande**, Professor & Head of Department, APCOER Pune (SPPU) · *Undergraduate Research Advisor (AI & Machine Learning)* · Contact available upon request

• **Prof. Sunil Thakare**, Senior Faculty & Academic Mentor, APCOER Pune (SPPU) · *Mathematics & Foundations Mentor* · Contact available upon request